

Project objective

Central research question:

Can a vision-language model correctly classify forest vegetation change and generate explanations grounded in multimodal remote-sensing and climate evidence?

What we are building

- A **benchmark dataset** (not a new RS classifier)
- **366** matched before/after forest patches
- **Five-source evidence cube** per patch
- **Condition C** evaluation — VLM sees *all* sensors + masks + boxes
- Explanations tied to bounding boxes on 512×512 images

Study design

- **Site:** Kanha Tiger Reserve
- **T0:** Jan–Feb 2022 **T1:** Jan–Feb 2023
- **Patch size:** ~640 m
- **Classes:** stable / decline / increase
- **Privacy:** no lat/lon in VLM JSON

Overall project completion: ~52%

Module	Weight	Done	Bar	Status
1. Data engineering	35%	~95%		Essentially complete
2. Dataset packaging	10%	~90%		366 Condition C cards
3. Labels & eval protocol	15%	~25%		candidate_class only
4. Explanations (Ollama)	5%	~30%		366 fallback templates
5. VLM experiments	25%	~0%		Not started
6. Results & thesis	10%	~25%		Mid-review deck only
Total project	100%	~52%		

Key message

Data engineering + dataset packaging are done.

Remaining: labels/eval splits, optional Ollama captions, VLM experiments.

Condition C: every data source in each evidence card

Condition C only — VLM receives the complete multimodal card.

JSON block	Sensor	Contents per patch
images	Sentinel-2 RGB	T0 + T1 natural-color 512×512 JPG
sentinel2	Sentinel-2	10 optical bands + NDVI/EVI/NDMI/NBR
sentinel1	Sentinel-1 SAR	VV, VH (T0 + T1)
chirps	CHIRPS	Rainfall mm (T0 + T1)
era5_land	ERA5-Land	Temp + soil-moisture anomalies
dem	NASADEM	Elevation, slope, aspect
evidence	Derived	Δ NDVI, Δ NBR + Otsu masks
triplets	Spatial GT	Bbox + disturbance type + severity
explanation	Caption	Ground-truth text

Pipeline: Notebook 01 (export) → 02 (DIP/boxes) → 03 (dataset.jsonl).

Repository map: what to show live

`ecoexplain_vlm_dataset/`

- `dataset.jsonl` — open 1 record
- `data/preview_chips/` — visual QA
- `data/raw_tiffs/` — 15 files/patch
- `data/processed_images/` — JPGs + tensors
- `data/evidence_masks/` — Otsu masks
- `results/` — validation reports
- `notebooks/01,02,03` — pipeline

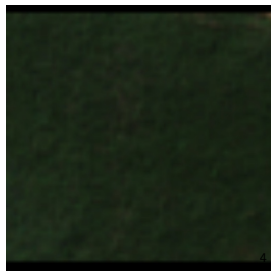
Demo order: JSONL → previews → notebook 02 → validation JSON.



T0 (2022)



T1 (2023) — Sentinel-2 RGB, not SAR



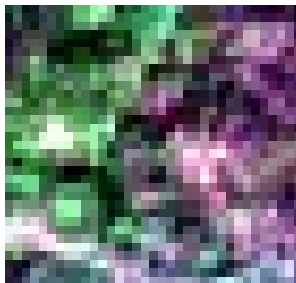
Purple-color issue, validation, and next steps

Why purple appeared

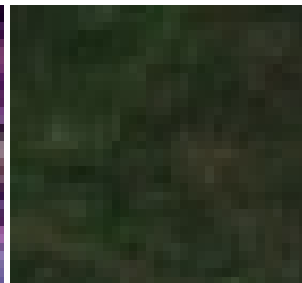
- **S2 RGB** chips, not VV/VH SAR
- Per-band stretch on small chips $\rightarrow$ false color
- **Fix:** fixed reflectance window ($v_{\max}=0.22$)

Validation

- 366/366 GeoTIFFs, 366 JSONL rows
- Preflight: **READY TO SCALE**



Wrong stretch (optical)



Correct (pipeline)

Next: train/dev/test splits $\rightarrow$ Condition C VLM experiments.